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Lab 07: Shared Library

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Objective

Extract a duplicated pipeline step into a versioned shared library consumed by two pipelines — reuse without copy-paste.

Prerequisites

- Lab 06 running; a second pipeline job; a git repo to host the library.

Architecture

Library repo (vars/notifySlack.groovy-style step) → Global Pipeline

Libraries → two Jenkinsfiles call the same step → one fix propagates to both.

Setup

Create repo pipeline-library with vars/standardTest.groovy:

```
def call(Map cfg = [:]) { sh "${cfg.cmd}?: 'npm test'}" }
```

Step 1 — Register the library

Execute

Manage Jenkins → System → Global Pipeline Libraries → add my-lib, default version main, source pointing at the repo.

Expected Result

Library listed, HEAD resolved.

Why

Versioned libraries are dependencies: pinned, reviewed, rollback-able — unlike copy-pasted steps.

Step 2 — Consume from two pipelines

Execute

In both Jenkinsfiles:

```
@Library('my-lib') _ pipeline { agent { label 'docker-agent' } stages { stage('Test') { steps { standardTest(cmd: 'npm test') } } } }
```

Expected Result

Both pipelines green, both calling the shared step.

Why

One definition, N consumers — the DRY payoff, with the version pin as the safety rail.

Verification

- [] Both pipelines call the library step.
- [] Breaking the library breaks both (proves sharing); reverting fixes both.
- [] Library change went through PR review on its own repo.

Failure Scenarios

- Library `my-lib` not found: name mismatch or version unresolvable — check Global config + repo branch.
- Untrusted library on sandbox: approve signatures or restrict to trusted source.

Cleanup

Keep the library repo — it is the seed of pipeline governance.

What This Proved

Reusable pipeline code with versioning. Continue with Lab 08.

Interview Notes

- Library not found: the two causes (name vs version) and where each is configured.
- Why library changes go through PR review on their own repo — what unreviewed shared code risks.

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60 min · INT

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